Exercise 8 - System Specifications

1 Introduction

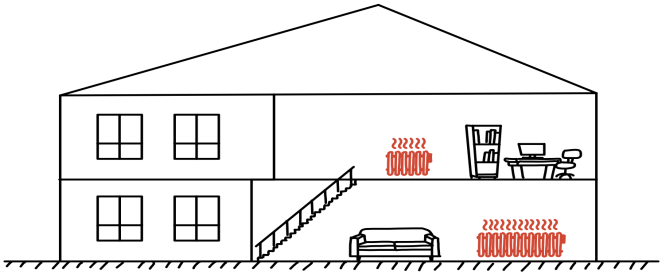
Last week we saw a method called Root Locus to establish whether the closed-loop system is stable or not, based on the open-loop transfer function (i.e. the open-loop poles and zeros). However, stability is not the only requirement for a control system. In fact, we also want the system to meet certain performance specifications and evaluate how well the closed-loop will behave.
For example, we may want to see how quickly and accurately the closed-loop system tracks commands, or how well it rejects disturbances, modeling errors and noise.

Today, we will focus on time-domain specifications (which are performance criteria defined in the time domain) and steady-state error (which is a measure of how accurately the system tracks reference inputs in the long term).

2 Time-Domain Specifications

2.1 First-Order System

To better understand the concept of time-domain specifications, let's consider a cold house that we want to heat up.



Assume that we can model the house as a first-order system:

$$G(s) = \frac{1}{\tau s + 1} \quad \Longleftrightarrow \quad \begin{cases} \dot{x} = -\frac{1}{\tau}x + \frac{1}{\tau}u \\ y = x \end{cases}$$

where u is the temperature set by the heater, and $y = x$ is the house temperature.

Imagine that we enter the house and set the heater to a temperature of $u(t) = 20°C$ (i.e. we want the house to reach a temperature of $20°C$). Furthermore, we assume that the initial temperature of the house is $x(0) = x_0 = 0°C$.
This corresponds to a scaled step response of the system:

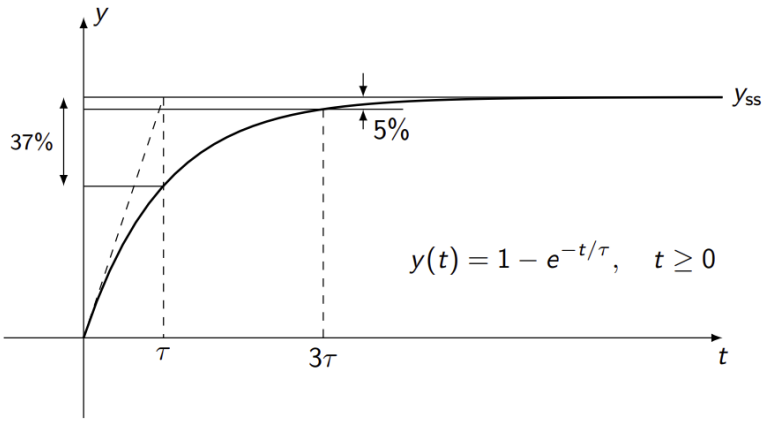
$$y(t) = \mathrm{C}e^{\mathrm{A}t}\cancel{x_0} + \mathrm{C}\int_0^t e^{\mathrm{A}(t-p)}\mathrm{B}u(p)dp + \cancel{D}u(t) =$$

$$\Downarrow$$

$$= 20\int_0^t \frac{1}{\tau}e^{-\frac{1}{\tau}(t-p)}dp =$$

$$\Downarrow$$

$$= 20\left(1 - e^{-\frac{t}{\tau}}\right)$$



We can see that the behavior of the system depends on the parameter τ, also called the *time constant*. In fact, the smaller τ is, the faster the system reaches the desired temperature.
We can use this to define the settling time T_d, i.e. the time it takes for the system to reach and stay within a certain percentage $d\%$ of the final value.

$$T_d = \tau \ln\left(\frac{100}{d}\right)$$

The settling time T_d can help us adjust the behavior of the system, such that it doesn't "grow" too quickly or too slowly. Furthermore, since T_d is proportional to τ, and τ influences the poles of $G(s)$, this time-domain specification translates into a pole location requirement in the complex plane.
Finally, this means that we can choose our controller parameters in such a way that the time-domain specifications are satisfied, by ensuring that the closed-loop poles are located in the appropriate region of the complex plane.

2.2 Second-Order System

However, usually the first-order system is not sufficient to model real-world systems. Therefore, let's approximate our situation from earlier by a second-order system. We can represent the heating system as

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \qquad \Longleftrightarrow \qquad \begin{cases} \dot{x} = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} x + \begin{bmatrix} 0 \\ \omega_n^2 \end{bmatrix} u \\ y = x_1 \end{cases}$$

where ω_n is the natural frequency and ζ is the damping ratio.

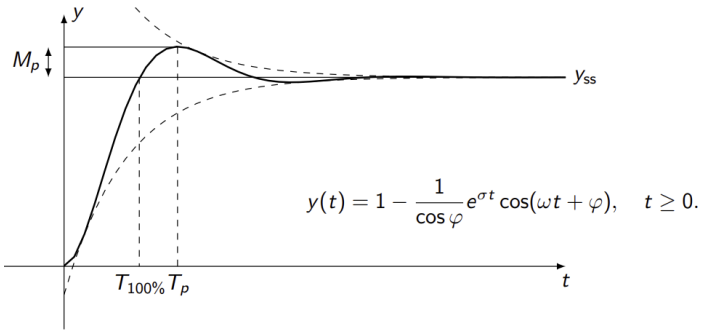
If we assume an underdamped system (i.e. $0 < \zeta < 1$) with zero initial conditions, the step response of the system is given by

$$y(t) = 1 - \frac{1}{\cos\varphi} e^{\sigma t} \cos(\omega t + \varphi), \qquad t \geq 0$$

with $\sigma = -\zeta\omega_n$, $\omega = \omega_n\sqrt{1-\zeta^2}$ and $\varphi = \arctan\left(\frac{\sigma}{\omega}\right)$.

Here the time constant is now $\tau = \frac{1}{|\sigma|}$. Therefore the settling time within a certain percentage $d\%$ of the steady-state for a second-order system is given by:

$$T_d = \tau \ln\left(\frac{100}{d}\right) = \frac{1}{|\sigma|} \ln\left(\frac{100}{d}\right)$$



We can also define other time-domain specifications, such as:

- **Time to peak** (Peak Time) T_p: the time it takes for the system to reach the first peak of the overshoot.

$$T_p = \frac{\pi}{\omega}$$

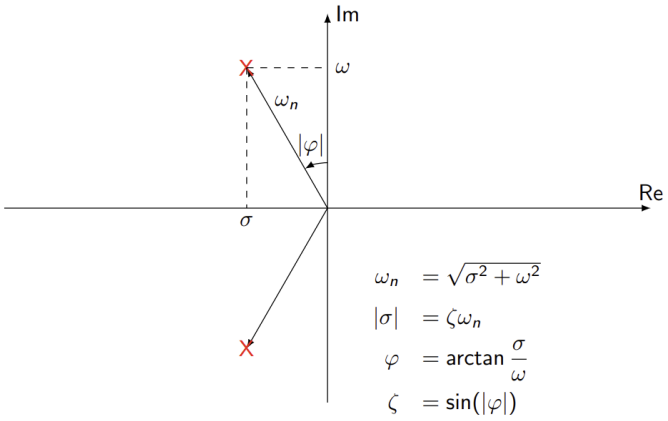
- **Peak Overshoot** (ratio) M_p: the maximum amount by which the response exceeds its steady-state value.

$$M_p = e^{\frac{\sigma\pi}{\omega}} \qquad \Leftrightarrow \qquad \zeta^2 = \frac{(\ln(M_p))^2}{\pi^2 + (\ln(M_p))^2}$$

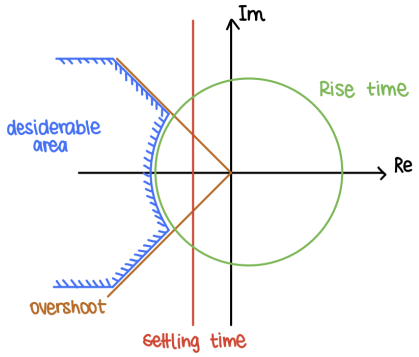
- **Rise Time** $T_{100\%}$: the time it takes for the system to go from 0% to 100% of the final value for the first time.

$$T_{100\%} = \frac{\frac{\pi}{2} - \varphi}{\omega} \approx \frac{\pi}{2\omega_n}$$

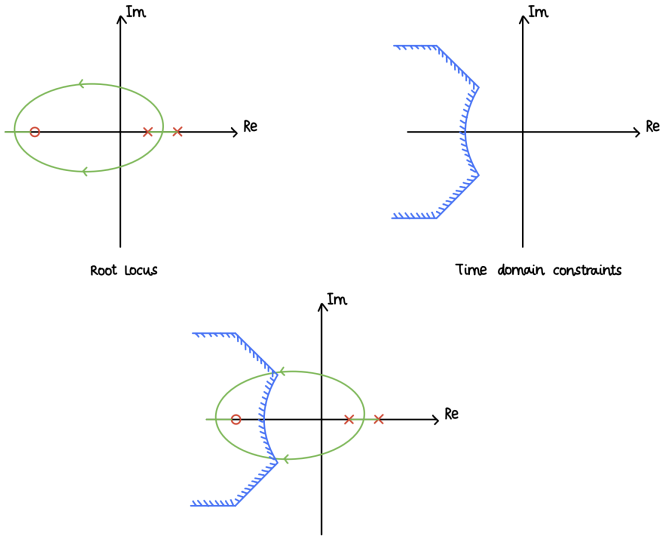
We can map the poles in the complex plane and observe how their locations determine the different components of the time response.



Therefore, we can again take the time-domain requirements and translate them into constraints on the pole locations in the complex plane.
This allows us to design our closed-loop system to meet the desired specifications by appropriately placing the closed-loop poles.



A way to relate these restrictions to the root locus is to overlay the two.

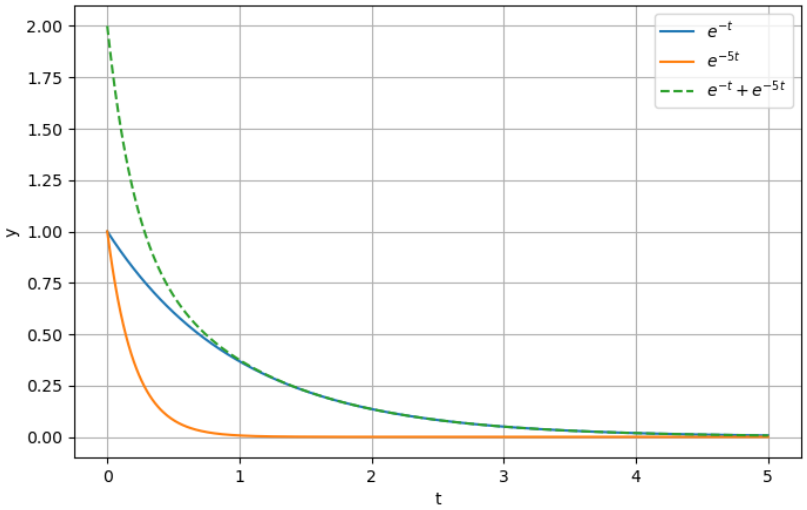


We can then choose the gain k such that the closed-loop poles (i.e. the points on the root locus) satisfy the time-domain specifications.

2.3 Dominant pole approximation

In many practical situations the system can be of higher order, i.e. order greater than two. In such cases it is often useful to approximate it with a second-order system, and apply the time-domain specifications accordingly.

Remember that the poles are related to an exponential term in the time domain. In addition, the real part of the pole determines how fast the exponential decays. Therefore, if we have a system with multiple poles, the poles with the largest real part (i.e. closest to the imaginary axis) will dominate the system's behavior, since their corresponding exponentials decay more slowly.



In fact, as shown in the figure above, the system with the pole further to the left will decay faster, while the one with the pole closer to the imaginary axis will decay slower. Finally, we observe that the system with both poles will behave similarly to the one with the pole closer to the imaginary axis, since its exponential term will dominate the overall response.
This is called the **dominant pole approximation**.

Example

We consider the following transfer function:

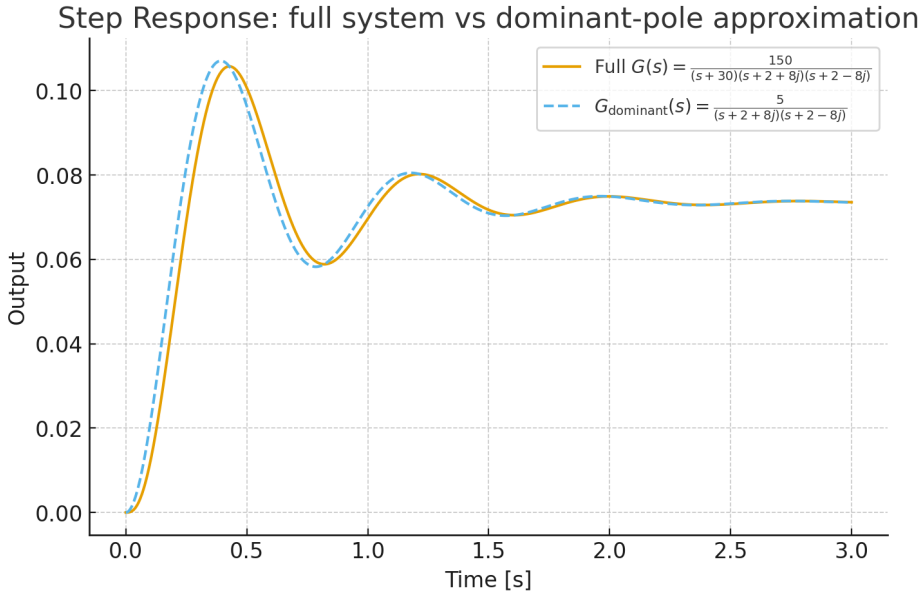
$$G(s) = \frac{150}{(s+30)(s+2+8j)(s+2-8j)}$$

The poles of the system are located at $s_1 = -30$, $s_2 = -2+8j$ and $s_3 = -2-8j$.

Additionally, the poles s_2 and s_3 are closer to the imaginary axis, and therefore they will dominate the system's behavior.
We can approximate the system as a second-order system (thus performing the dominant pole approximation) by neglecting the "faster" pole at $s_1 = -30$.
Therefore, we have:

$$G_{dominant}(s) = \frac{150}{30}\frac{1}{(s+2+8j)(s+2-8j)} = \frac{5}{(s+2+8j)(s+2-8j)}$$

Note: The factor $\frac{1}{30}$ is introduced to ensure that the steady-state value of the approximated system matches that of the original system.



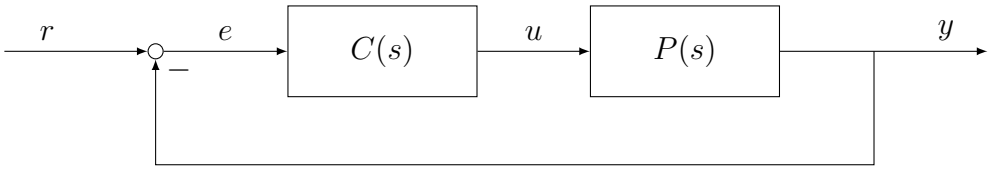
Conclusion: To perform the dominant pole approximation, we neglett the most negative poles from the transfer function, adjusting the gain to maintain the same steady-state value.
Important note: The dominant pole approximation could also be performed by a pole-zero cancellation, if there are zeros very close to the poles.

3 Steady-State Error

In Exercise 6, we saw the Final Value Theorem

$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} s\,Y(s) = \lim_{s\to 0} s\,G(s)U(s)$$

This theorem allows us to compute the steady-state value of the output.
Now, if we consider a closed-loop system,



we can also define a transfer function from r to e as

$$S(s) = \frac{E(s)}{R(s)} = \frac{1}{1 + P(s)C(s)} = \frac{1}{1 + L(s)}$$

This function is called the **sensitivity function**, and it describes how the error signal $e(t)$ responds to changes in the reference input $r(t)$.

Using the Final Value Theorem, we can now compute the steady-state error as

$$e_{ss} = \lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) = \lim_{s\to 0} sS(s)R(s) = \lim_{s\to 0} s\frac{1}{1+L(s)}R(s)$$

Additionally, using a unit step input $r(t) = 1$, we have $R(s) = \frac{1}{s}$, and therefore we get

$$e_{ss} = \lim_{s\to 0} s\frac{1}{1+L(s)}\frac{1}{s} = \lim_{s\to 0}\frac{1}{1+L(s)} = \frac{1}{1+L(0)}$$

Furthermore, if we look at $L(s)$ in Bode form we can see that

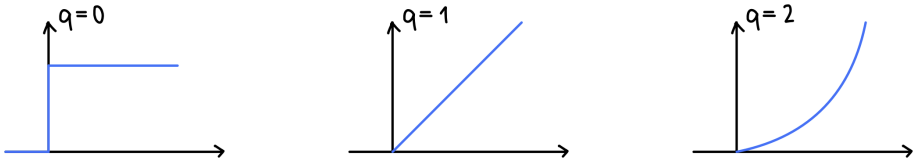
$$L(s) = \frac{k_{bode}}{s^{n_I}}\frac{\left(\frac{s}{-z_1}+1\right)\dots\left(\frac{s}{-z_m}+1\right)}{\left(\frac{s}{-p_1}+1\right)\dots\left(\frac{s}{-p_{n-n_I}}+1\right)}$$

$$\lim_{s\to 0} L(s) = \lim_{s\to 0}\frac{k_{bode}}{s^{n_I}} = \begin{cases} k_{bode} & \text{if } n_I = 0 \\ \infty & \text{if } n_I \geq 1 \end{cases}$$

where n_I is the number of poles at the origin, i.e. the type of system.

Therefore, we can conclude that if the closed-loop system is stable and there are a sufficient number of poles at the origin (also called *integrators*), **the steady-state error will be zero!**
The required number of integrators depends on the input signal.
Normally, when looking at the steady-state error we consider ramp inputs and these are defined as

$$r(t) = \frac{1}{q!}t^q \quad \Longleftrightarrow \quad R(s) = \frac{1}{s^{q+1}}$$



To summarize everything we said so far, depending on the **type of system** (i.e. number of integrators) and the **order of the ramp**, we can have different steady-state errors:

e_{ss}	q=0	q=1	q=2
Type 0	$\frac{1}{1+k_{bode}}$	∞	∞
Type 1	0	$\frac{1}{k_{bode}}$	∞
Type 2	0	0	$\frac{1}{k_{bode}}$

To conclude, the steady-state error can be reduced by increasing the Bode gain k_{bode} and if we want to have a zero steady-state error to ramps of order q, it is necessary to have at least $q+1$ integrators.

Example

Given an open-loop transfer function $L(s) = \frac{s+8}{s^2+5s}$, we want to compute the steady-state error to a 1st order ramp.

First, we can write $L(s)$ in Bode form:

$$L(s) = \frac{8}{5} \cdot \frac{1}{s} \frac{\frac{s}{8}+1}{\frac{s}{5}+1} \quad \text{, where} \quad k_{bode} = \frac{8}{5}, \text{ and} \quad n_I = 1$$

Since we have only a single integrator, this is a Type 1 system.
Finally, using the table from earlier, we can see that the steady-state error is given by

$$e_{ss} = \frac{1}{k_{bode}} = \frac{5}{8}$$

Important note: The Final Value Theorem would give the same result.

Exam Problem: HS24

Problem: Consider the closed-loop system T shown in Figure 10, where P is a linear time-invariant plant, and $C_i, i = 1, 2, 3, 4, 5$ are PD controllers.

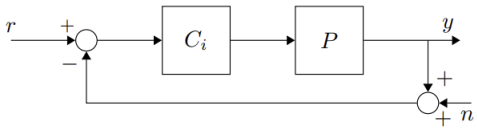


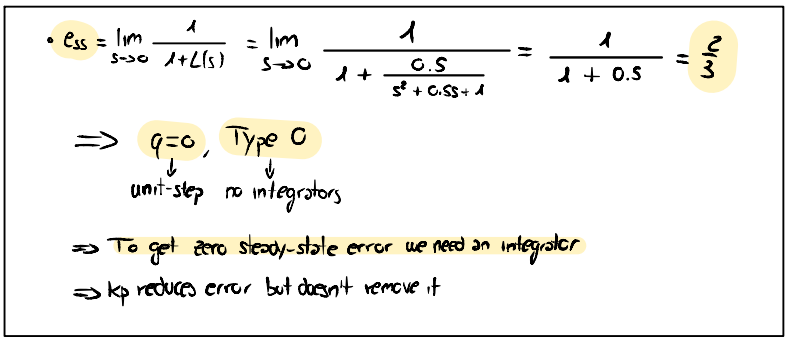
Figure 10: Standard feedback architecture. LTI plant P and PD controllers $C_i, i = 1, 2, 3, 4, 5$.

The transfer function of the plant P is given by:

$$P(s) = \frac{0.5}{s^2 + 0.5s + 1}.$$

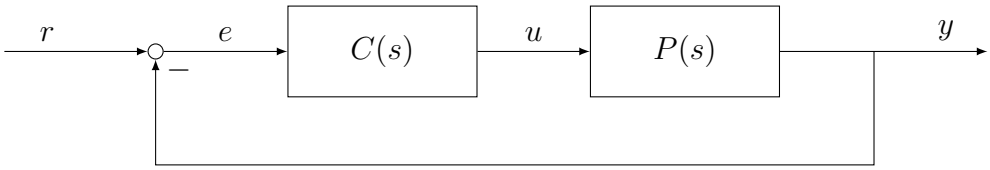
Q24 (0.75 Points) Mark all correct statements.

- [A] To a unit-step reference, the closed-loop system T with $C_5(s) = 1$ has zero steady state error.
- To a unit-step reference, the closed-loop system T with $C_5(s) = 1$ has a steady state error of $\frac{2}{3}$.
- To ensure that the closed-loop system has no steady state error, the controller should include an integrator.
- [D] To ensure that the closed-loop system has no steady state error, the proportional gain of the controller can be set to a very large value.



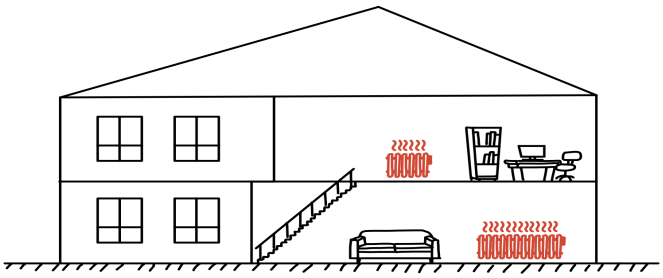
4 PID Control

Recall our closed-loop system and what our controller does:



The controller takes the error, i.e. the difference between the desired reference and the current output of the system, and transforms it into an input for the plant.
In other words, the controller's goal is to convert $e(t)$ into an input $u(t)$ for the plant.

To understand this better, let's consider a heated room.



In this situation the reference $r(t)$ is the desired temperature T_{ref}, while the output $y(t)$ is the measured temperature T. Thus, the error is $e(t) = T_{\text{ref}} - T$.
The controller interprets this error and generates an input, for example, the power delivered to the heating element.
In other words, it translates *"how far we are from the target"* into *"how much we need to heat"*.
But how do we convert the error into a command?

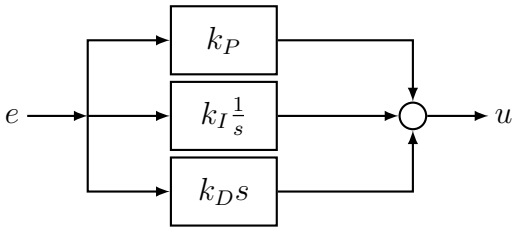
One way to do it is with a **PID controller!** What does PID stand for?

P → Proportional

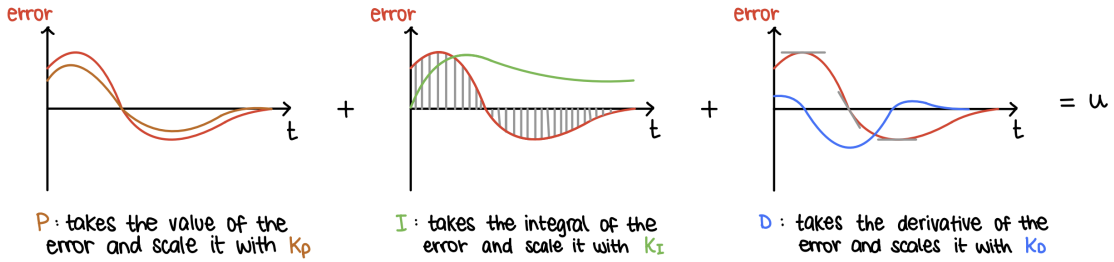
I → Integral

D → Derivative

each term handles the error differently



But how do we interpret what each term does?



Here, the transfer function of the PID controller is given by

$$C(s) = k_P + \frac{k_I}{s} + k_D s = \frac{k_D s^2 + k_P s + k_I}{s}$$

or in the time domain:

$$u(t) = k_P\, e(t) + k_I \int_0^t e(\tau) d\tau + k_D\, \dot{e}(t)$$

Therefore, the challenge is to choose the parameters k_P, k_I and k_D in such a way that the closed-loop system meets the desired specifications.
This is called **PID tuning**.

Prof. Frazzoli's PID tuning method:

1. Assume proportional control **P**.
2. Draw the root locus.
3. If the root locus does NOT go through the "good" region ⇒ Need to add a **D** term. Go back to step 2.
4. Choose a gain that places the dominant poles in the "good" region.
5. If the steady-state error is too large ⇒ Need to add a **I** term. Go back to step 2.
6. If **PID** is not enough, stay tuned for more advanced methods.

The effects of each term can be summarized as follows:

- **Proportional term**:
  - Produces a faster response.
  - Decreases steady-state error.
  - Increases closed-loop bandwidth.
  - Decreases disturbance sensitivity.
  - Increases sensitivity to noise.
  - Can reduce stability margin for high-order systems.
- **Integral term**:
  - Eliminates steady-state error (as long as the closed-loop system is stable).
  - Reduces stability margins.
  - Response becomes slower.
  - Can introduce overshoot and oscillations.
- **Derivative term**:
  - Reduces overshoot and oscillations.
  - Improves stability margins.
  - Increases sensitivity to noise.

Exam Problem: FS23

Problem: Consider the standard feedback architecture shown in Figure 14 where a PID-controller $C(s) = k_P + k_D s + k_I/s$ is used to control a linear time-invariant system P. Further, assume that P can be described by a strictly rational transfer function with no poles or zeros at $s = 0$.

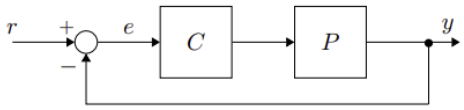


Figure 14: Standard feedback architecture. System P and controller C.

Q29 **(0.5 Points)** For this question consider the integral and derivative gains, k_I and k_D respectively, fixed. Mark the correct answer for each statement.

Statement	True	False
Increasing or decreasing the proportional gain k_P can change the phase margin of the closed-loop system.	X	
Let $k_I = 0$. Increasing the proportional gain k_P can decrease the steady-state error of the closed-loop system to a step response.	X	
Increasing the proportional gain k_P can increase the bandwidth of the closed-loop system and make it respond faster.	X	

Q30 **(0.5 Points)** For this question consider the proportional and derivative gains, k_P and k_D respectively, fixed. Mark the correct answer for each statement.

Statement	True	False
A zero steady-state error to a step reference is achieved if the integral gain k_I is zero.		X
When slightly increasing the integral gain k_I, the sensitivity of the closed-loop system to high-frequency noise increases considerably.		X
Increasing the integral gain k_I can decrease the phase margin of the closed-loop system and can cause the closed-loop system to become unstable.	X	

Q31 **(0.5 Points)** For this question consider the proportional and integral gains, k_P and k_I respectively, fixed. Mark the correct answer for each statement.

Statement	True	False
Introducing a non-zero derivative gain k_D can help to reduce overshoot.	X	
Introducing a non-zero derivative gain k_D can help increase the phase margin.	X	
Introducing a non-zero derivative gain k_D does not change the steady-state error.	X	

See summary of the effects of each term!

Exam Problem: HS23

Problem: Consider the standard feedback architecture shown in Figure 19 where a PID-controller $C(s) = k_\mathrm{P} + k_\mathrm{D}s + k_\mathrm{I}/s$ is used to control a linear time-invariant system P. Further, assume that P can be described by a strictly proper rational transfer function with no poles or zeros at $s = 0$.

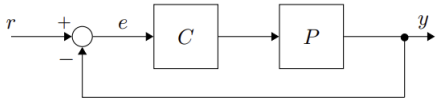


Figure 19: Standard feedback architecture. System P and controller C.

Q35 **(1 Points)** Mark all correct statements.

- [x] Assume k_I and k_D fixed. Increasing k_P can make the system respond faster.
- [x] Assume k_D and k_P fixed. In order to guarantee a zero steady-state error $e_\mathrm{ss} = 0$ to step references, it is necessary that $k_\mathrm{I} \neq 0$.
- [C] Assume k_I and k_P fixed. By tuning k_D we can reduce the steady-state error e_ss to step references to zero.
- [x] Assume k_I and k_P fixed. By increasing k_D we can reduce the overshoot of the closed-loop step response.

See summary of the effects of each term!